

Backend Engineer Task:

Design and implement a simple program (GO / NodeJS) to find specific strings in a large text. The program should be composed of the following modules:

- The main module - reads a large text file in parts (e.g. 1000 lines in each part) and sends each part (as string) to a matcher. After all matchers completed, it calls the aggregator to combine and print the results
- The matcher - gets a text string/s as input and searches for matches of a given set of strings. The result is a map from a word to its location(s) in the text
- The aggregator - aggregates the results from all the matchers and prints the results

For this task, please use the text at <http://norvig.com/big.txt> (will act as a big file, consider it could be each file with each volume), and the strings to find should be the 50 most common English first names:

James, John, Robert, Michael, William, David, Richard, Charles, Joseph, Thomas, Christopher, Daniel, Paul, Mark, Donald, George, Kenneth, Steven, Edward, Brian, Ronald, Anthony, Kevin, Jason, Matthew, Gary, Timothy, Jose, Larry, Jeffrey, Frank, Scott, Eric, Stephen, Andrew, Raymond, Gregory, Joshua, Jerry, Dennis, Walter, Patrick, Peter, Harold, Douglas, Henry, Carl, Arthur, Ryan, Roger

Example of one line from the program output based on the input above:

Timothy --> [{lineOffset:13000, charOffset:19775}, {lineOffset:13000, charOffset:42023}]

Couple more things:

- There should be several concurrent matchers (i.e each matcher should run in a separate thread / worker / routine)
- The results should be printed (in no particular order) after all text pieces have been processed
- Please provide a main method that executes a sample run
- This task should be done alone. You can make use of any written or web resource you find, incl. 3rd party libraries
- The code should work, and be as clean and efficient as you think necessary

Good Luck!